# ITSI

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# TASK 1

We chose to analyze data set with information about influence of diseases and drugs on human’s energy and oxygen saturation.

```
library(ggplot2)
```

```
## Warning: pakiet 'ggplot2' został zbudowany w wersji R 4.4.2
```

```
library(car)
```

```
## Warning: pakiet 'car' został zbudowany w wersji R 4.4.2
```

```
## Ładowanie wymaganego pakietu: carData
```

```
## Warning: pakiet 'carData' został zbudowany w wersji R 4.4.2
```

```
library(fitdistrplus)
```

```
## Warning: pakiet 'fitdistrplus' został zbudowany w wersji R 4.4.3
```

```
## Ładowanie wymaganego pakietu: MASS
```

```
## Ładowanie wymaganego pakietu: survival
```

```
library(ggpubr)
```

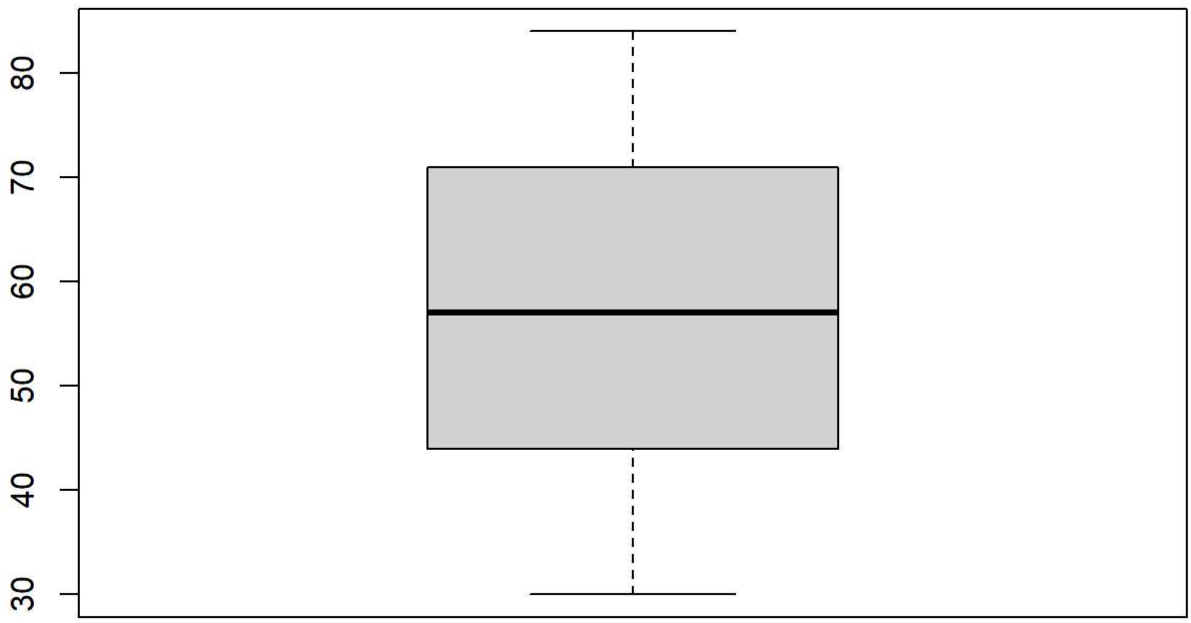
```
## Warning: pakiet 'ggpubr' został zbudowany w wersji R 4.4.3
```

```
sample= read.csv(file="C:/Users/sus/Desktop/project ITSI/Lung Cancer Dataset.csv")
sample=sample[!is.na(sample$GENDER), ]
sample$ALCOHOL_CONSUMPTION = ifelse(sample$ALCOHOL_CONSUMPTION == 1, "YES", "NO")
head(sample)
```

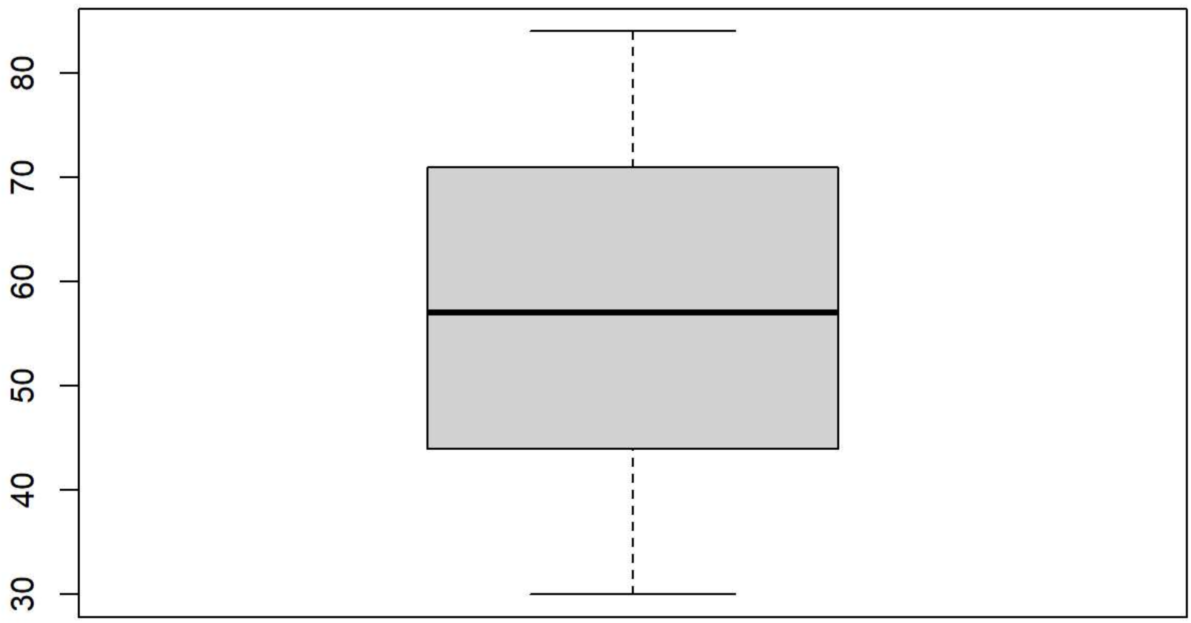
```
##   AGE GENDER SMOKING FINGER_DISCOLORATION MENTAL_STRESS EXPOSURE_TO_POLLUTION
## 1  68      1       1                    1             1                     1
## 2  81      1       1                    0             0                     1
## 3  58      1       1                    0             0                     0
## 4  44      0       1                    0             1                     1
## 5  72      0       1                    1             1                     1
## 6  37      1       1                    1             1                     1
##   LONG_TERM_ILLNESS ENERGY_LEVEL IMMUNE_WEAKNESS BREATHING_ISSUE
## 1                 0     57.83118               0               0
## 2                 1     47.69484               1               1
## 3                 0     59.57744               0               1
## 4                 0     59.78577               0               1
## 5                 1     59.73394               0               1
## 6                 1     57.68429               0               1
##   ALCOHOL_CONSUMPTION THROAT_DISCOMFORT OXYGEN_SATURATION CHEST_TIGHTNESS
## 1                 YES                 1          95.97729               1
## 2                  NO                 1          97.18448               0
## 3                 YES                 0          94.97494               0
## 4                  NO                 1          95.18790               0
## 5                  NO                 1          93.50301               0
## 6                 YES                 1          94.05715               1
##   FAMILY_HISTORY SMOKING_FAMILY_HISTORY STRESS_IMMUNE PULMONARY_DISEASE
## 1              0                      0             0                NO
## 2              0                      0             0               YES
## 3              0                      0             0                NO
## 4              0                      0             0               YES
## 5              0                      0             0               YES
## 6              0                      0             0               YES
```

We formulated problem. We want to know if there exist age interval with cumulation of people that drink alcohol.

```
drunk = sample$AGE[sample$ALCOHOL_CONSUMPTION=="YES"]
sober = sample$AGE[sample$ALCOHOL_CONSUMPTION=="NO"]
boxplot(sober)
```

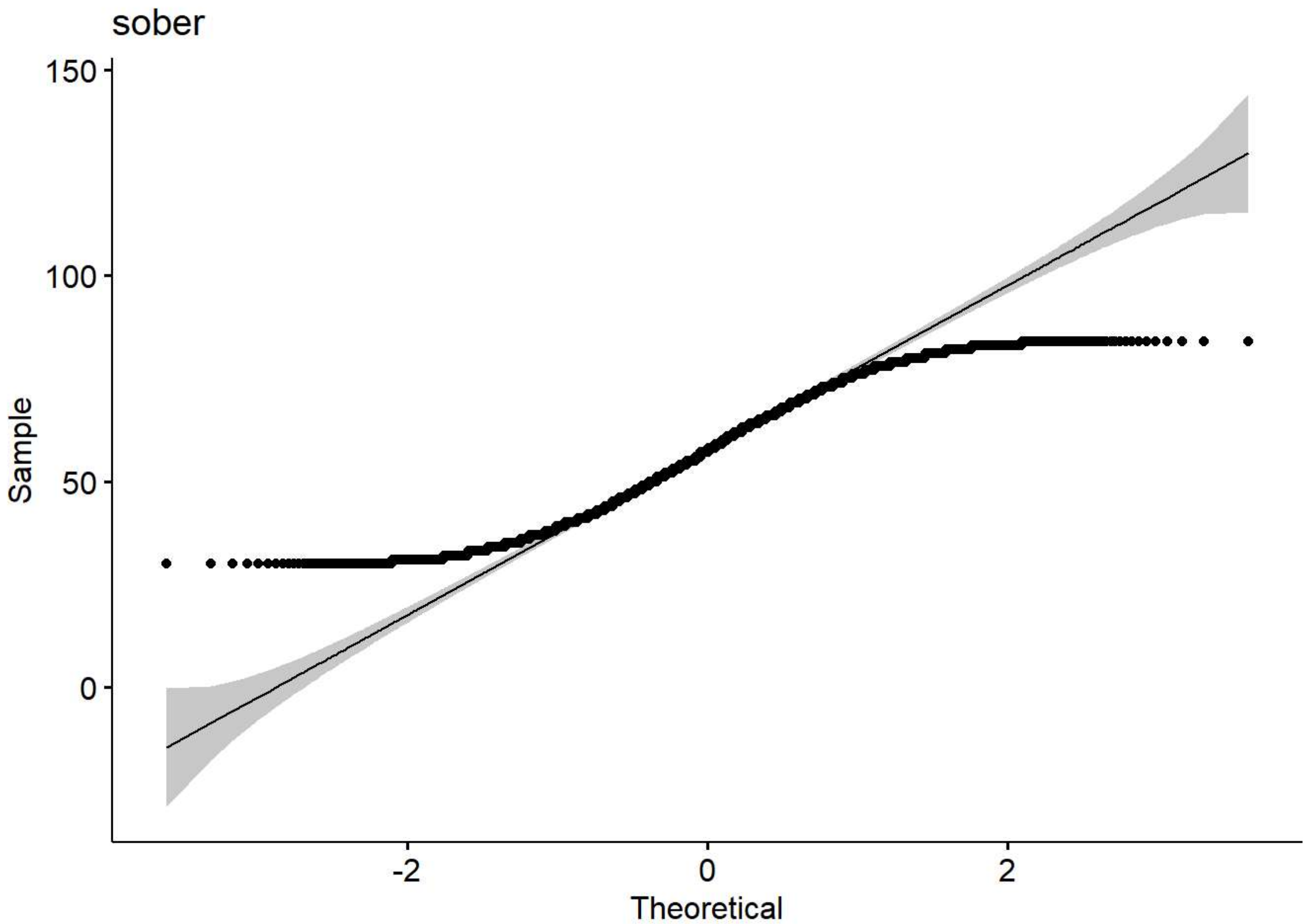



```
boxplot(drunk)
```

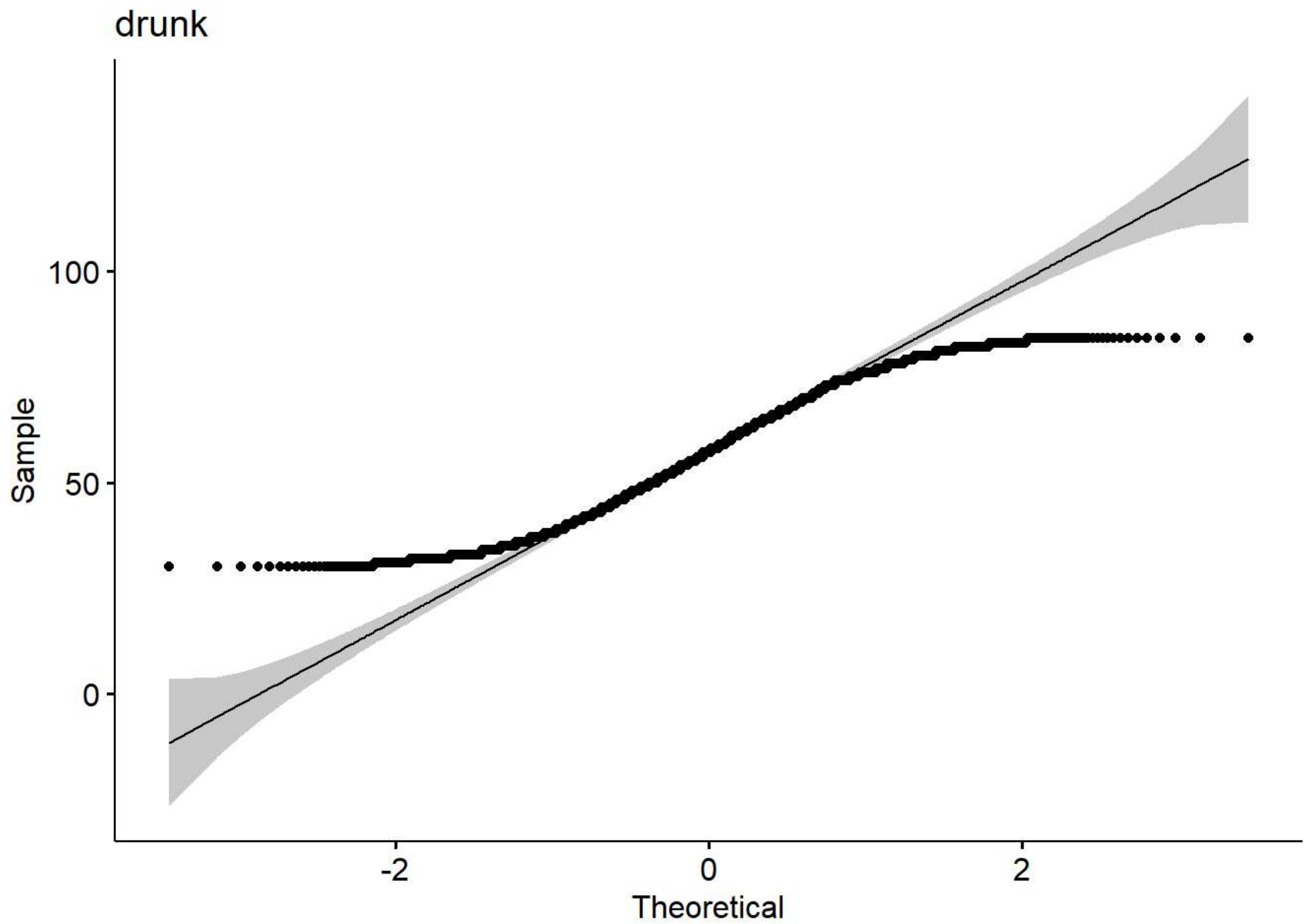


From boxplots, we can see that there aren’t any outliers for both samples, so we don’t have to clean them from data set.

```
ggqqplot(sober, conf.int = TRUE, conf.int.level = 0.95, title = "sober")
```

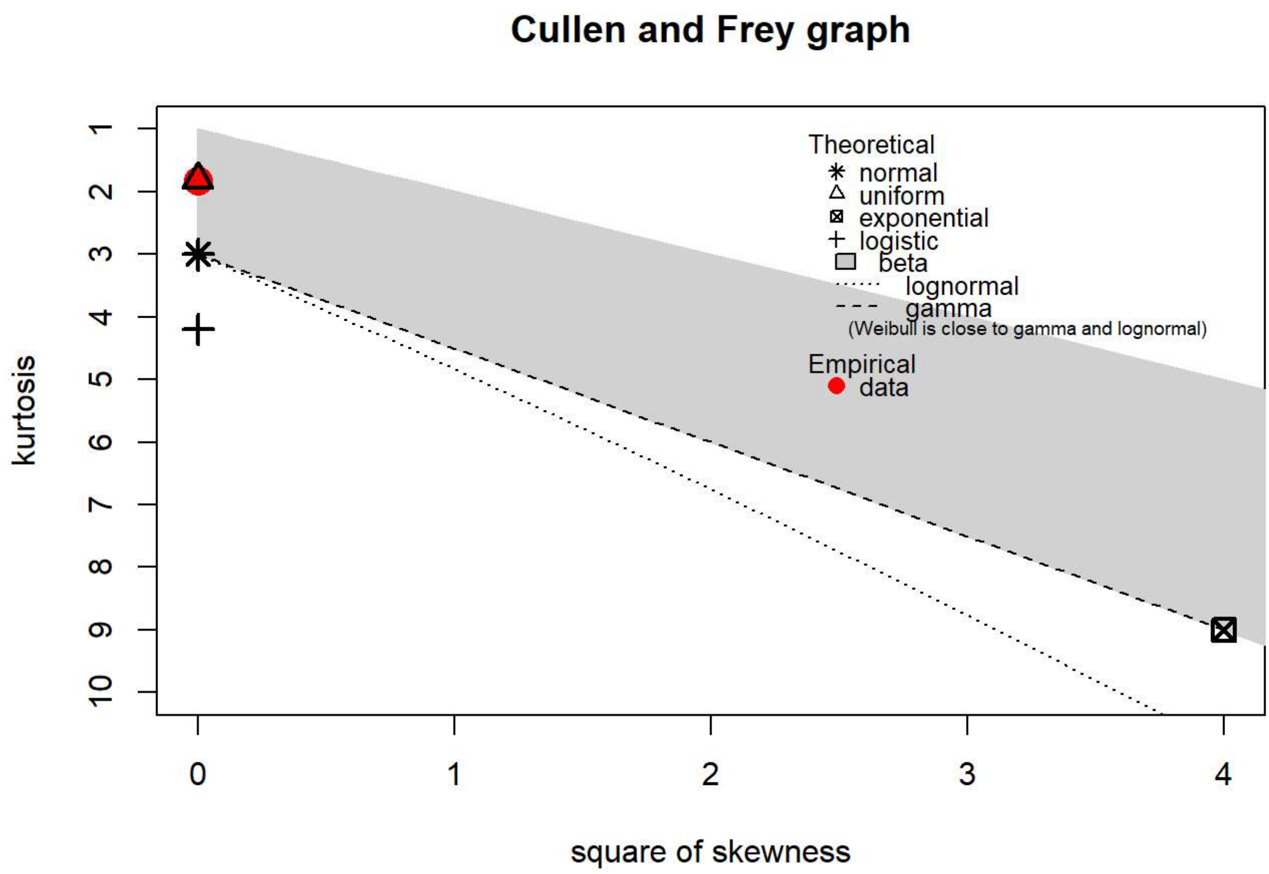



```
ggqqplot(drunk, conf.int = TRUE, conf.int.level = 0.95, title = "drunk")
```



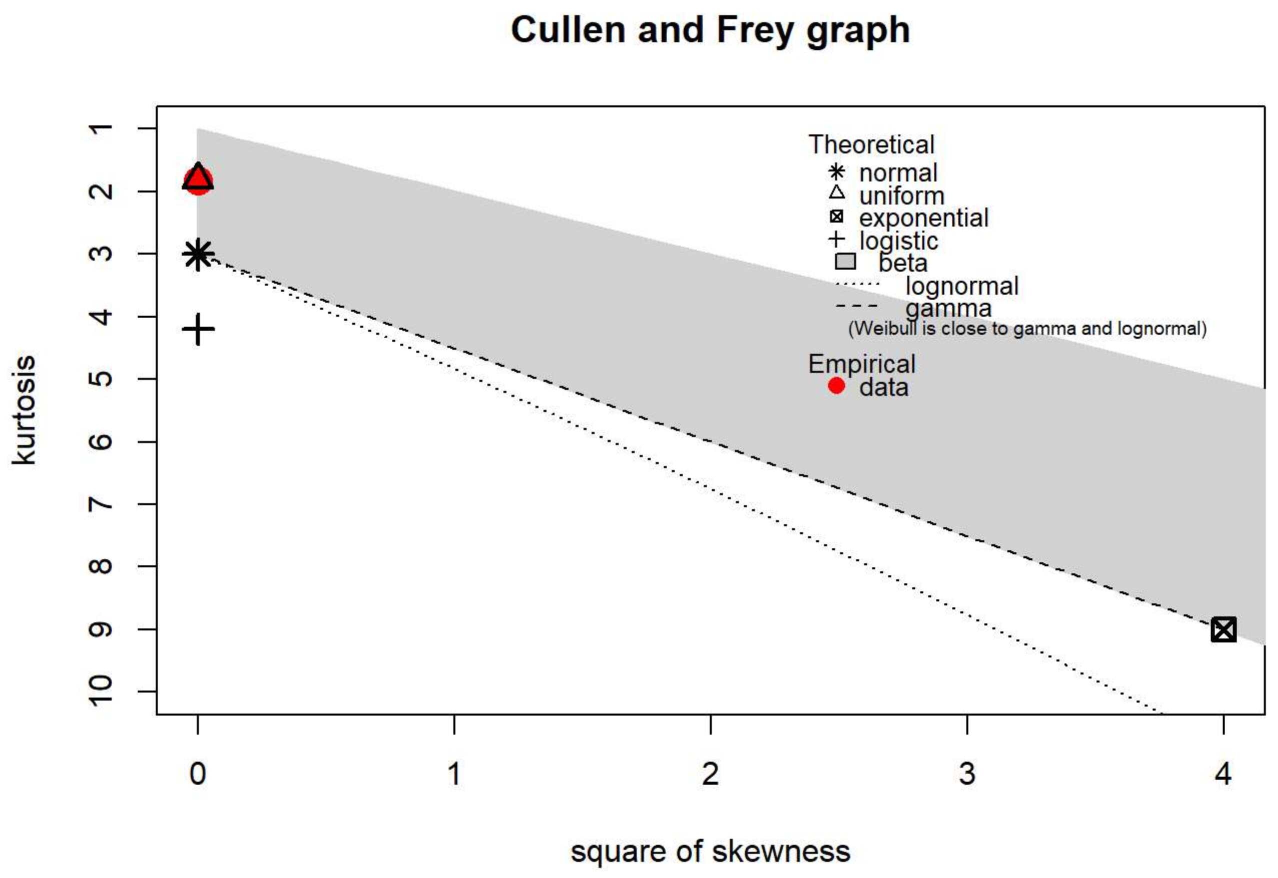

Ggplots show that distribution of our sample is most likely not normal. Now we want to find what is their PDF.

```
descdist(sober)
```




```
## summary statistics
## ------
## min:  30   max:  84
## median:  57
## mean:  57.24094
## estimated sd:  15.79467
## estimated skewness:  -0.0212469
## estimated kurtosis:  1.825445
```

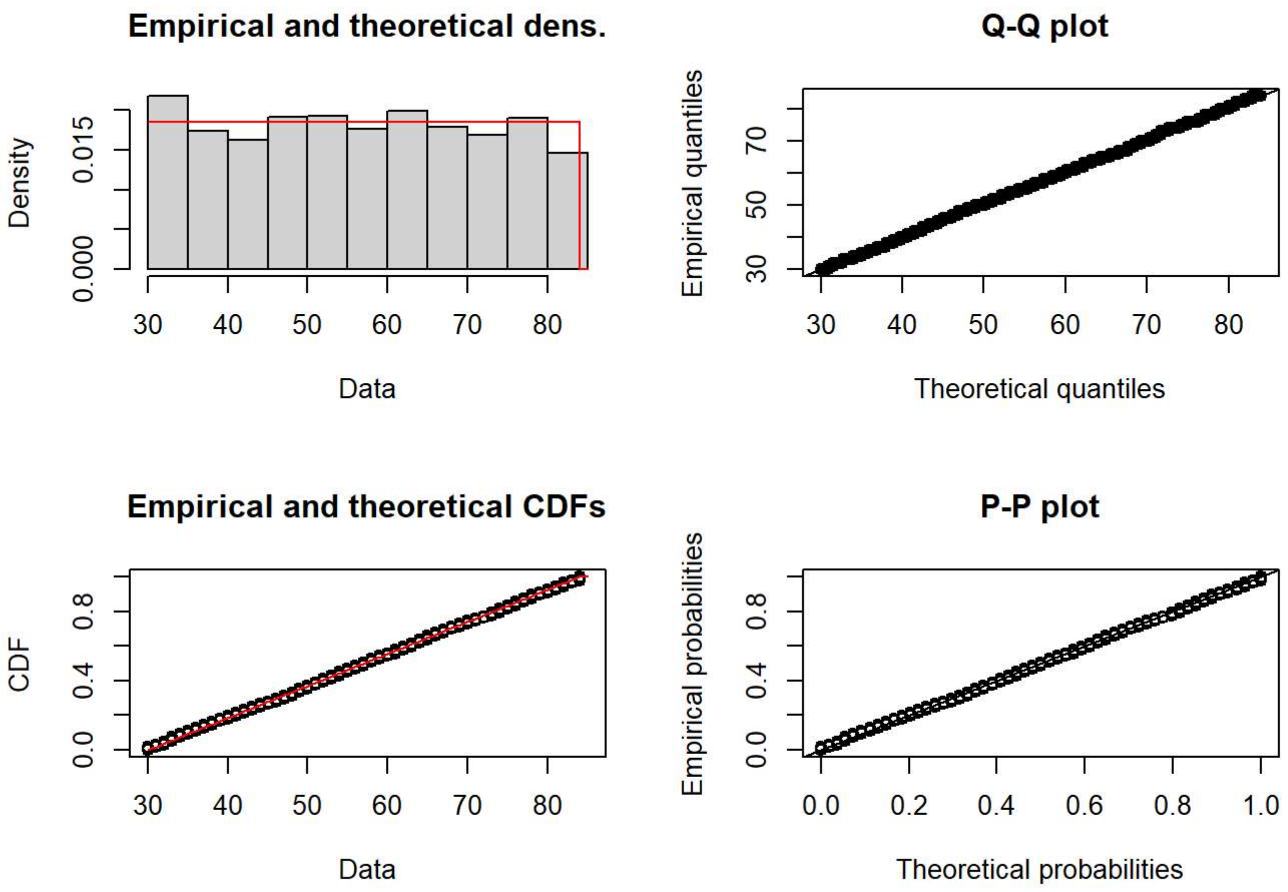
```
descdist(drunk)
```




```
## summary statistics
## ------
## min:  30   max:  84
## median:  57
## mean:  57.18972
## estimated sd:  15.81193
## estimated skewness:  -0.01285026
## estimated kurtosis:  1.820157
```

From Cullen and Frey graph we clearly see that we deal with uniform distribution, but to be sure let’s check it with different plots, and Shapiro-Wilk test.

```
fit_uni <-fitdist(drunk, "unif")
plot(fit_uni)
```




```
shapiro.test(drunk)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  drunk
## W = 0.95526, p-value < 2.2e-16
```

```
shapiro.test(sober)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  sober
## W = 0.95618, p-value < 2.2e-16
```

Now we are 100% sure that our sample distribution is uniform, because we reject hypothesis about normality of our samples. Now we want to check our start Hypothesis. We need to chose proper tests. First we check Homogeneity of variance. Our samples don't follow normal distribution, so we will use Levene's test.

```
mean(drunk)
```

```
## [1] 57.18972
```

```
mean(sober)
```

```
## [1] 57.24094
```

```
leveneTest(sample$AGE~sample$ALCOHOL_CONSUMPTION)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##         Df F value Pr(>F)
## group    1   2e-04 0.9885
##       4998
```

P-value of the test is equal to 0.9885, so we don't reject Null Hypothesis about equality of variances. To finalize our task, we will use Wilcoxon test.

```
wilcox.test(sober,drunk,paired = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  sober and drunk
## W = 2864700, p-value = 0.9116
## alternative hypothesis: true location shift is not equal to 0
```

P-value there is equal to 0.9116, so we also cannot reject hypothesis about equality of variances of samples. This leads us to conclusion that there don't exist any interval with highest number of people that drink alcohol. That may be caused by inappropriate sample.

# TASK 2

Our task is to compare powers of t-test and Wilcoxon test under normality samples. Only assumption for samples is to be independent, that is why we chose to generate samples with rnorm function in R language.

```
simulate_power = function(n, d, test_type = c("t", "wilcox"), conf = 0.1, iterations = 10000)
{
  test_type = match.arg(test_type)
  rejections = 0
  for (i in 1:iterations) {
    x = rnorm(n, mean = 0, sd = 1)
    y = rnorm(n, mean = d, sd = 1)
    if (test_type == "t") {
      p_val = t.test(x, y, var.equal = TRUE)$p.value
    } else {
      p_val = wilcox.test(x, y)$p.value
    }
    if (p_val < conf) {
      rejections = rejections + 1
    }
  }
  return(rejections / iterations)
}


set.seed(2137)


sample_sizes = c(25, 50)
effect_sizes = c(1/8, 1/4, 1/2)
for (n in sample_sizes) {
  for (d in effect_sizes) {
    power_t = simulate_power(n, d, "t")
    power_w = simulate_power(n, d, "wilcox")
    cat(sprintf("n = %d, d = %.3f | t-test power = %.3f, Wilcoxon power = %.3f\n", n, d, powe
r_t, power_w))
  }
}
```

```
## n = 25, d = 0.125 | t-test power = 0.130, Wilcoxon power = 0.133
## n = 25, d = 0.250 | t-test power = 0.221, Wilcoxon power = 0.214
## n = 25, d = 0.500 | t-test power = 0.536, Wilcoxon power = 0.513
## n = 50, d = 0.125 | t-test power = 0.169, Wilcoxon power = 0.164
## n = 50, d = 0.250 | t-test power = 0.350, Wilcoxon power = 0.339
## n = 50, d = 0.500 | t-test power = 0.801, Wilcoxon power = 0.779
```

We perform tests for samples x and y where x is normally distributed sample and y’s mean is changed as said in task, so its equal to half, quarter and 1/8 in each next loop. After simulating the tests we display our results.

Only thing that is left to complete, is to find number of samples, such that the estimation error is less than 0.005 with probability at least 99%.

```
z = qnorm(0.995)
p = 0.1
m = 0.005
n_required = (z^2 * p * (1 - p)) / (m^2)
n_required
```

```
## [1] 23885.63
```

Calculations give us number “23885.63”. Sample size needs to be integer, so after rounding it we get 23886. That finalize our project.